UNITED STATES PATENT APPLICATION

Application Type: Non-Provisional Utility Patent Application **Filed Under:** 35 U.S.C. § 111(a)

INVENTOR

Name: Erkin Ötleş, MD, PhD

TITLE OF THE INVENTION

SYSTEM AND METHOD FOR EXPERTISE-GATED PUBLICATION OF ARTIFICIAL INTELLIGENCE RELATED SOCIAL MEDIA CONTENT

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of and priority to U.S. Provisional Patent Application No. [PENDING], filed [DATE], the entire contents of which are incorporated herein by reference.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH

Not Applicable.

FIELD OF THE INVENTION

The present invention relates generally to social media content moderation systems, and more particularly to computer-implemented methods and systems for selectively gating the publication of social

media posts concerning artificial intelligence (AI) and related topics based on automated verification of the posting user's technical expertise in those domains.

BACKGROUND OF THE INVENTION

The proliferation of artificial intelligence (AI) technologies has been accompanied by an exponential increase in social media discourse concerning AI capabilities, limitations, societal implications, and future trajectories. A substantial portion of this discourse originates from individuals who lack foundational technical knowledge of the underlying systems they describe, resulting in widespread dissemination of inaccurate claims, exaggerated capabilities, unfounded predictions, and misleading characterizations of AI technologies.

This phenomenon—commonly referred to as "AI hype"—imposes measurable social costs. Non-expert AI commentary distorts public understanding of AI capabilities, influences policy discussions with technically unsound premises, shapes investment and career decisions based on false information, and crowds out substantive expert discourse. The cumulative effect is an information environment in which technically accurate, nuanced AI commentary is epistemically disadvantaged relative to sensationalized, technically erroneous content that more readily attracts engagement on social media platforms.

Existing content moderation systems address harmful content categories such as hate speech, disinformation, spam, and illegal material. However, no prior art system addresses the specific problem of AI domain expertise verification as a prerequisite to publication. Existing systems that require credential verification (e.g., professional networks requiring employment history) are passive, asynchronous, and do not perform real-time, technically validated expertise verification at the point of content creation.

Existing CAPTCHA-style systems verify that a user is human rather than automated, using perceptual tasks (image recognition, text transcription) that are specifically designed to be trivial for humans.

No prior art system employs a CAPTCHA-analogous mechanism that selectively challenges users based on detected domain-specific content and validates domain expertise through domain-specific technical problems.

Accordingly, there exists a need in the art for a system and method that (1) detects AI-related content in social media posts prior to publication, (2) presents the posting user with a technical domain-expertise challenge analogous to a CAPTCHA, (3) evaluates the user's response to determine domain competency, and (4) selectively permits or prevents publication based on that determination.

SUMMARY OF THE INVENTION

The present invention provides a computer-implemented system and method for expertise-gated publication of artificial intelligence related social media content. In one aspect, the invention intercepts a social media post prior to publication, analyzes the post for AI-related content using natural language processing and/or keyword matching techniques, and, upon detecting AI-related content, presents the user with a technical expertise verification challenge. The challenge comprises a problem drawn from a domain associated with artificial intelligence, such as linear algebra, machine learning theory, or AI programming. The user's response is evaluated against a correctness criterion, and publication is authorized or denied accordingly.

In a first aspect, the invention provides a computer-implemented method for regulating publication of social media content. The method comprises receiving a social media post submitted by a user; analyzing the post to detect AI-related content; upon detection, generating and presenting a technical expertise challenge; evaluating a user response to the challenge; and selectively permitting or preventing publication based on the evaluation outcome.

In a second aspect, the invention provides a system comprising one or more processors and memory storing instructions that implement the method described herein, including an AI content analyzer module, a challenge generator module, a response evaluator module, and a post disposition module.

In a third aspect, the invention provides a non-transitory computer-readable medium storing instructions implementing the foregoing method.

The present invention may optionally associate a verified expertise indicator (e.g., a "Verified AI Expert" badge) with posts published by users who successfully complete the expertise verification challenge. Posts that fail the challenge may be saved as drafts and accompanied by curated educational resources to encourage the development of genuine AI expertise.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention.

FIG. 1 is a block diagram of the system architecture according to an embodiment, illustrating the relationship among the Social Media Platform Interface, AI Content Analyzer, Challenge Generator, Response Evaluator, and Post Disposition modules.

FIG. 2 is a flowchart depicting the computer-implemented method according to a preferred embodiment, including the decision branches for AI content detection, challenge presentation, response evaluation, and post disposition.

FIG. 3 is a block diagram of the Challenge Generator subsystem, illustrating the Problem Bank, Difficulty Adjuster, Category Selector, and Problem Formatter components.

FIG. 4 is a series of user interface state diagrams showing (A) the post composition state, (B) the content analysis state with animated keyword detection, (C) the expertise challenge presentation state, (D) the publication success state with verified expert indicator, and (E) the publication failure state with draft saving and educational resources.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The following detailed description sets forth specific embodiments of the invention. Those of skill in the art will recognize that numerous modifications and substitutions may be made without departing from the spirit and scope of the claims appended hereto.

I. System Overview

Referring to FIG. 1, the system 100 comprises a Social Media Platform Interface 110, an AI Content Analyzer 120, a Challenge Generator 130, a Response Evaluator 140, a Post Disposition Module 150, and a User Interface Manager 160, all implemented on one or more computing devices comprising at least one processor and non-transitory computer-readable memory.

The Social Media Platform Interface 110 is configured to receive social media posts submitted by users prior to publication and to transmit publication commands to an underlying social media platform or to withhold such commands based on instructions from the Post Disposition Module 150.

The AI Content Analyzer 120 is configured to receive the text content of a social media post and to determine whether that content contains AI-related material. In a preferred embodiment, the AI Content Analyzer 120 employs a multi-tier analysis approach comprising: (a) keyword and phrase matching against a curated lexicon of AI-related terms including, without limitation, terms such as "artificial intelligence," "machine learning," "deep learning," "neural network," "large language model," "GPT," "transformer architecture," "diffusion model," "reinforcement learning," "AGI," and "generative AI"; (b) optionally, semantic analysis using a language model to identify AI-related claims that may not contain explicit keywords; and (c) a confidence scoring mechanism that determines whether the detected AI content meets a publication-gating threshold.

The Challenge Generator 130 is configured to generate, upon receiving a trigger from the AI Content Analyzer 120, an expertise verification challenge comprising a technical problem drawn from a domain associated with AI. In a preferred embodiment, the Challenge

Generator 130 maintains a Problem Bank 131 comprising a plurality of technical problems organized by category and difficulty. Problem categories include, without limitation: (a) linear algebra problems including determinant calculation, eigenvalue identification, matrix rank determination, and vector norm computation; (b) computer programming problems including code output prediction, algorithm complexity analysis, and AI framework API usage; (c) machine learning theory problems including attention mechanism properties, activation function behavior, and optimization algorithm characteristics; and (d) statistics and probability problems relevant to AI systems.

The Response Evaluator 140 is configured to receive a user's response to the expertise verification challenge and to determine whether the response satisfies a correctness criterion. In embodiments employing multiple-choice challenges, the correctness criterion is satisfied when the user selects the predetermined correct answer. In embodiments employing open-ended challenges, the Response Evaluator 140 may employ a language model or rule-based natural language processing to assess correctness.

The Post Disposition Module 150 is configured to instruct the Social Media Platform Interface 110 to (a) publish the post if the Response Evaluator 140 determines the response is correct, or (b) prevent publication if the response is incorrect, with the option to save the post as a draft accessible to the user.

The User Interface Manager 160 is configured to render the successive interface states described in connection with FIG. 4 and to handle user interactions including text entry, answer selection, challenge refresh requests, and post reset actions.

II. Method Description

Referring to FIG. 2, the computer-implemented method 200 proceeds as follows.

Step 210 — Post Reception. The system receives a social media post comprising text content submitted by a user for publication on a social media platform. In a preferred embodiment, this step is triggered

when the user activates a "Post" or equivalent publication control in the user interface.

Step 220 — AI Content Analysis. The AI Content Analyzer 120 analyzes the text content of the received post. In a preferred embodiment, analysis proceeds by iterating through the AI keyword lexicon and identifying matches within the post text, with optional additional semantic analysis. The system computes an AI-content score based on the number and significance of detected AI-related terms and claims.

Step 230 — Content Classification Decision. The system determines whether the AI-content score meets or exceeds a predetermined gating threshold. If the score does not meet the threshold (i.e., no significant AI content is detected), the method proceeds to Step 260. If the score meets or exceeds the threshold, the method proceeds to Step 240.

Step 240 — Challenge Generation and Presentation. Upon determining that the post contains AI-related content, the Challenge Generator 130 selects a technical problem from the Problem Bank 131 and presents it to the user via the User Interface Manager 160. The user interface displays (a) a notification that AI content has been detected; (b) the detected AI keywords found in the post; (c) the technical challenge problem, which may include a mathematical expression, matrix representation, code snippet, or descriptive question; (d) a plurality of selectable answer options (in the multiple-choice embodiment); (e) a submission control; and (f) a challenge refresh control enabling the user to request an alternative problem.

Step 250 — Response Reception and Evaluation. The system receives the user's response via the user interface and the Response Evaluator 140 determines whether the response is correct. If correct, the method proceeds to Step 260. If incorrect, the method proceeds to Step 270.

Step 260 — Post Publication. The Post Disposition Module 150 instructs the Social Media Platform Interface 110 to publish the post. In embodiments in which the user has completed a successful expertise verification challenge, the User Interface Manager 160 associates a verified expertise indicator (e.g., a "✓ AI Expert" badge)

with the published post. The user is presented with a success notification.

Step 270 — Post Prevented. The Post Disposition Module 150 prevents publication of the post. In a preferred embodiment, the post is saved as a draft accessible to the user for future editing and re-submission. The user is presented with an encouraging failure notification indicating that further development of AI expertise is encouraged. The user interface optionally presents curated educational resources relevant to AI and machine learning, including references to publicly available courses, textbooks, and instructional video series.

III. Challenge Generator Subsystem

Referring to FIG. 3, the Challenge Generator 130 comprises a Problem Bank 131, a Category Selector 132, a Difficulty Adjuster 133, and a Problem Formatter 134.

The Problem Bank 131 stores a plurality of technical problems, each associated with metadata including category, difficulty level, problem text, optional supplementary content (e.g., a matrix representation or code snippet), answer options, correct answer, and an explanation used for educational feedback upon failure.

The Category Selector 132 optionally biases problem selection toward categories relevant to the detected AI content in the user's post. For example, a post discussing transformer architectures may trigger a machine learning theory problem concerning attention mechanisms, while a post concerning AI-generated images may trigger a problem concerning diffusion models or convolutional architectures.

The Difficulty Adjuster 133, in embodiments maintaining a per-user expertise profile, adjusts the difficulty of selected problems based on the user's history of challenge attempts and outcomes. Users who have previously succeeded on basic problems may be presented with more advanced problems in subsequent sessions.

The Problem Formatter 134 renders the selected problem in a form suitable for display in the user interface, including rendering matrix notation, code blocks with syntax-appropriate formatting, and mathematical expressions.

IV. Alternative Embodiments

Embodiment A — Language Model–Assisted Content Analysis. In an alternative embodiment, the AI Content Analyzer 120 submits the post text to an AI language model (e.g., a large language model accessible via an API) with a prompt instructing the model to determine whether the post makes AI-related claims and, if so, to identify the specific claims and their technical accuracy. This embodiment provides richer semantic understanding beyond keyword matching and can identify AI-related claims expressed without standard terminology.

Embodiment B — Language Model–Generated Challenges. In an alternative embodiment, the Challenge Generator 130 uses a language model to dynamically generate technical challenges tailored to the specific AI topic of the detected post, rather than drawing from a static Problem Bank. This embodiment provides greater variety and topical relevance at the cost of requiring backend language model infrastructure.

Embodiment C — Tiered Verification. In an alternative embodiment, users may accumulate a verified expertise status over time by successfully completing a required number of challenges across multiple posting sessions, reducing or eliminating the challenge requirement for subsequent AI-related posts during a defined verification period.

Embodiment D — Integration with Professional Credentials. In an alternative embodiment, the system supplements technical challenge verification with optional linkage to external professional credential systems (e.g., academic publications, professional certifications, or institutional affiliations) to provide additional pathways to expertise verification.

Embodiment E — Open-Ended Challenge Responses. In an alternative embodiment, challenges may require free-text rather than multiple-choice responses, with the Response Evaluator 140 employing a language model or structured natural language processing to evaluate correctness and partial credit.

V. User Interface State Descriptions

Referring to FIG. 4, the user interface transitions through the following states:

(A) Composition State. The user is presented with a text entry area, a character counter, and a post submission control. Example post prompts may be provided to demonstrate system behavior.

(B) Analysis State. Upon submission, the system displays the post text in a preview panel alongside an animated content scan. AI-related keywords detected in the post are progressively highlighted, providing the user with immediate visual feedback on the content triggering the verification requirement.

(C) Challenge State. If AI content is detected, the interface transitions to the challenge state, displaying the flagged post text with detected keywords, the technical problem with its supplementary content, selectable answer options, a submission control, and a challenge refresh control.

(D) Success State. Upon correct challenge response, the interface displays the published post with a "✓ AI Expert" verified expertise indicator, the correct answer explanation, and a control to compose a new post.

(E) Failure State. Upon incorrect challenge response, the interface displays an encouraging message, a draft-saved notification, the correct answer and explanation for educational purposes, curated learning resources, and controls to view the saved draft or retry the challenge.

CLAIMS

What is claimed is:

1. A computer-implemented method for regulating publication of social media content concerning artificial intelligence, the method comprising: - receiving, at one or more processors, a social media post submitted by a user for publication on a social media platform, the social media post comprising text content; - analyzing, by the one or more processors, the text content to determine whether the social

media post contains artificial intelligence related content, the analyzing comprising at least one of: keyword matching against a lexicon of artificial intelligence related terms, phrase matching against a corpus of artificial intelligence related expressions, or semantic analysis of the text content; - upon determining that the social media post contains artificial intelligence related content, generating, by the one or more processors, an expertise verification challenge, the expertise verification challenge comprising a technical problem selected from a domain associated with artificial intelligence technology; - presenting, via a user interface, the expertise verification challenge to the user; - receiving, via the user interface, a response from the user to the expertise verification challenge; - evaluating, by the one or more processors, the response to determine whether the response satisfies a predetermined correctness criterion; - upon determining that the response satisfies the correctness criterion, authorizing publication of the social media post on the social media platform; and - upon determining that the response fails to satisfy the correctness criterion, preventing publication of the social media post.

2. The method of claim 1, wherein the domain is selected from the group consisting of: linear algebra, machine learning theory, neural network architecture, computer programming in artificial intelligence frameworks, statistics as applied to machine learning, and optimization theory.

3. The method of claim 1, wherein the technical problem comprises a multiple-choice question and wherein evaluating the response comprises determining whether the user selected a predetermined correct answer from among a plurality of presented answer options.

4. The method of claim 1, wherein generating the expertise verification challenge further comprises selecting the technical problem from a stored problem bank comprising a plurality of pre-authored technical problems organized by category and difficulty level.

5. The method of claim 1, further comprising: - presenting the user with a control to request an alternative expertise verification challenge; and - upon receiving a request for an alternative challenge

via the user interface, generating and presenting a different technical problem as a replacement challenge.

6. The method of claim 1, wherein preventing publication of the social media post comprises saving the social media post as a draft post retrievable by the user for subsequent editing and re-submission.

7. The method of claim 1, further comprising: - upon preventing publication of the social media post, presenting the user with one or more educational resources related to artificial intelligence topics.

8. The method of claim 1, further comprising: - upon preventing publication of the social media post, presenting an encouraging message to the user, the message recommending further development of technical expertise in artificial intelligence prior to contributing to artificial intelligence discourse.

9. The method of claim 1, further comprising: - upon authorizing publication of the social media post, associating a verified expertise indicator with the published post, the verified expertise indicator being visually distinguishable in the social media platform's post display.

10. The method of claim 1, wherein analyzing the text content comprises invoking a large language model to classify the text content as artificial intelligence related or non-artificial intelligence related and to identify specific artificial intelligence related claims present in the text.

11. The method of claim 1, further comprising: - maintaining, in computer-readable memory, a per-user expertise profile recording a history of expertise verification attempts and outcomes for the user; and - adjusting the difficulty of subsequently generated expertise verification challenges based on the per-user expertise profile.

12. The method of claim 1, further comprising, during the analyzing step, presenting to the user via the user interface a visual representation of the analysis process including progressive display of detected artificial intelligence related keywords found in the text content of the social media post.

13. The method of claim 1, wherein the expertise verification challenge comprises a supplementary content block comprising at

least one of: a matrix representation of a linear algebra problem, a code snippet in a programming language associated with artificial intelligence development, or a mathematical expression rendered for display.

14. A system for regulating publication of social media content concerning artificial intelligence, the system comprising: - one or more processors; and - a non-transitory computer-readable memory storing instructions that, when executed by the one or more processors, cause the system to: - receive a social media post submitted by a user for publication; - analyze the text content of the social media post to determine whether the post contains artificial intelligence related content; - upon determining that the post contains artificial intelligence related content, generate and present to the user an expertise verification challenge comprising a technical problem in a domain associated with artificial intelligence; - receive and evaluate a user response to the expertise verification challenge; and - selectively authorize or prevent publication of the social media post based on the evaluation.

15. The system of claim 14, wherein the memory further stores a problem bank comprising a plurality of technical problems each associated with a category, a difficulty level, answer options, a correct answer, and an explanatory description, and wherein generating the expertise verification challenge comprises selecting a problem from the problem bank.

16. The system of claim 14, wherein the memory further stores instructions that cause the system to: - save the social media post as a draft when publication is prevented; and - present the user with at least one educational resource and a disclosure of the correct answer to the expertise verification challenge.

17. The system of claim 14, wherein the memory further stores instructions that cause the system to associate a verified expertise indicator with the social media post upon authorizing publication following successful completion of the expertise verification challenge.

18. The system of claim 14, wherein the memory further stores a content analyzer module configured to perform keyword matching

against a stored lexicon of artificial intelligence related terms and to compute a content score based on the number and significance of matched terms.

19. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the processors to perform a method comprising: - receiving a social media post intended for publication; - determining whether the social media post contains content related to artificial intelligence by analyzing the post for artificial intelligence related keywords, phrases, or semantic content; - if the social media post is determined to contain artificial intelligence related content: - generating an expertise verification challenge comprising a technical problem selected from a domain associated with artificial intelligence; - presenting the expertise verification challenge to the user via a user interface; - receiving and evaluating a user response to the expertise verification challenge; - if the response is correct, authorizing publication of the social media post and associating a verified expertise indicator with the published post; and - if the response is incorrect, preventing publication of the social media post, saving the post as a draft, and presenting the user with an encouraging message and curated educational resources related to artificial intelligence.

20. The non-transitory computer-readable medium of claim 19, wherein the method further comprises presenting the user, prior to evaluating the response, with a control enabling the user to request an alternative expertise verification challenge, and upon receiving such a request, generating and presenting a replacement technical problem.

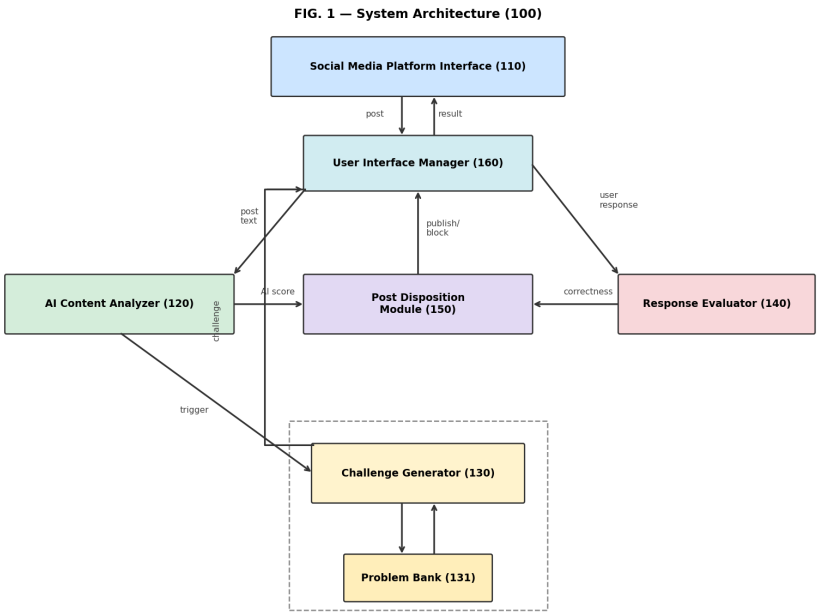
ABSTRACT

A computer-implemented system and method for expertise-gated publication of social media content concerning artificial intelligence (AI). Upon a user's attempt to publish a social media post, the system analyzes the post text to detect AI-related content using keyword matching, phrase detection, and/or semantic analysis. Posts identified as containing AI-related content trigger an expertise verification challenge—analogous to a CAPTCHA—comprising a technical problem drawn from an AI-related domain such as linear algebra,

machine learning theory, or AI programming. The user may answer the challenge or request an alternative problem. If the user responds correctly, the post is published and may carry a verified expertise indicator. If the user responds incorrectly, publication is prevented, the post is saved as a draft, and the user receives an encouraging message together with curated AI educational resources. The invention reduces the propagation of technically inaccurate AI commentary on social media platforms by requiring demonstrated domain expertise as a precondition to AI-topic publication.

BRIEF DESCRIPTION OF DRAWINGS (FIGURE DESCRIPTIONS)

FIG. 1 — System Architecture

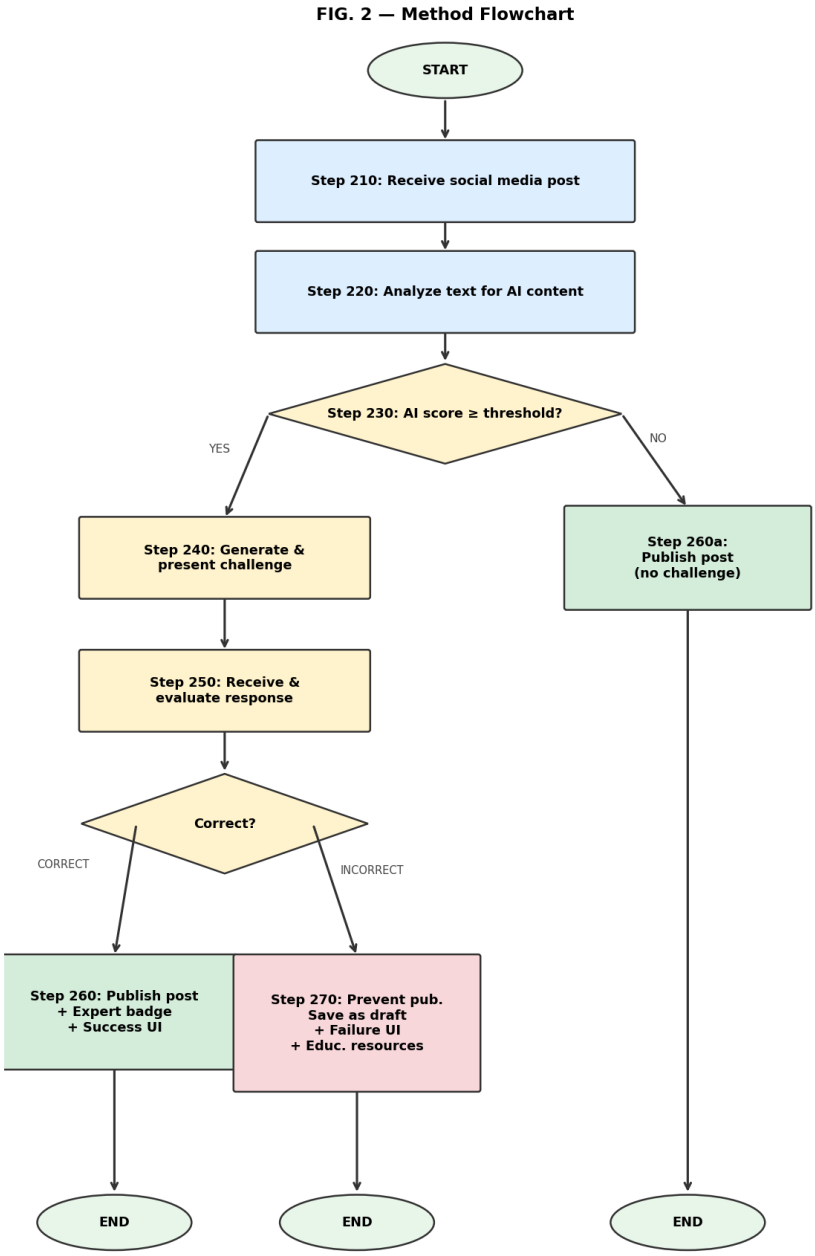


A block diagram showing system 100 with the following components connected by directional arrows indicating data flow:

- **Social Media Platform Interface (110):** Receives user post; issues publish/block commands to underlying platform.
- **AI Content Analyzer (120):** Receives post text from 110; computes AI-content score; triggers Challenge Generator if threshold is met; communicates score to Post Disposition Module.

- **Challenge Generator (130):** Receives trigger from 120; accesses Problem Bank 131; outputs challenge to User Interface Manager 160.
- **Problem Bank (131):** Stored repository of technical problems, sub-component of 130.
- **Response Evaluator (140):** Receives user response from User Interface Manager 160; determines correctness; reports to Post Disposition Module 150.
- **Post Disposition Module (150):** Receives correctness determination from 140; issues publish or block instruction to Social Media Platform Interface 110; instructs User Interface Manager 160 to display success or failure state.
- **User Interface Manager (160):** Renders all interface states; mediates user interactions; forwards user response to Response Evaluator 140.

FIG. 2 — Method Flowchart

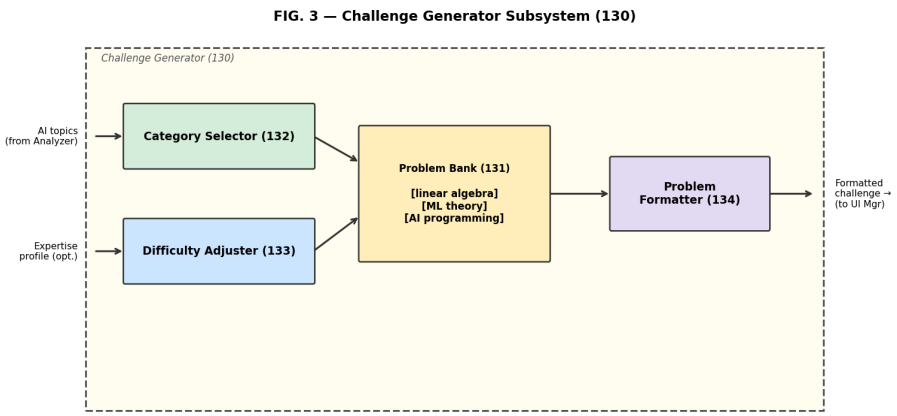


A flowchart with the following nodes and decision branches:

```
[START]
    |
    ▼
[Step 210] Receive social media post from user
    |
    ▼
```

```
[Step 220] Analyze text for AI-related content
    |
    ▼
[Step 230] AI content score ≥ threshold?
   / YES              NO \
  ▼                       ▼
[Step 240]         [Step 260a]
Generate &         Publish post
present            (no challenge)
challenge              |
  |                    ▼
  ▼                  [END]
[Step 250]
Receive &
evaluate response
  / CORRECT      INCORRECT \
 ▼                          ▼
[Step 260]             [Step 270]
Publish post           Prevent publication
+ Expert badge         Save as draft
+ Success UI           + Failure UI
+ Explanation          + Correct answer
     |                 + Educ. resources
     ▼                       |
   [END]                     ▼
                           [END]
```

FIG. 3 — Challenge Generator Subsystem

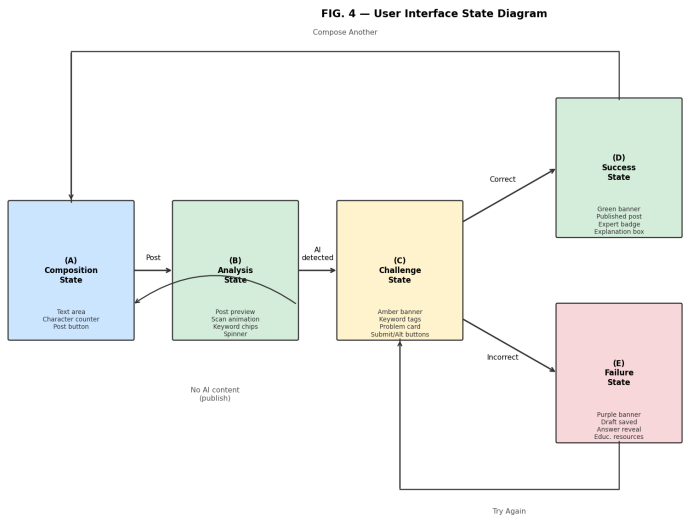


A block diagram of the Challenge Generator (130) showing:

- **Category Selector (132):** Receives detected AI topics from AI Content Analyzer; selects appropriate problem category.

- **Difficulty Adjuster (133):** Optionally receives user expertise profile; selects difficulty level.
- **Problem Bank (131):** Multi-category repository queried by Category Selector and Difficulty Adjuster.
- **Problem Formatter (134):** Receives raw problem from Problem Bank; renders mathematical notation, code blocks, and answer options; outputs formatted problem to User Interface Manager.

FIG. 4 — User Interface State Diagram



Five labeled interface states:

(A) Composition State: Text area with placeholder, character counter, example post suggestion pills, Post button.

(B) Analysis State: Post text preview box with animated scan line; row of keyword chips that highlight in red when matched; spinner and "Analyzing content…" label.

(C) Challenge State: Amber warning banner ("AI Expertise Verification Required"); flagged post preview with highlighted keywords; keyword tag row; problem card with category badge, question text, optional code/math block, multiple-choice option list; Submit Answer button and Different Problem button.

(D) Success State: Green banner ("Expertise Verified!"); published post card with "✓ AI Expert" badge; explanation box with correct-answer rationale; Compose Another Post button.

(E) Failure State: Purple banner ("Keep Learning!"); encouraging descriptive text; draft-saved indicator with post snippet; yellow answer-reveal box with correct answer and explanation; learning resources list; View Draft and Try Again buttons.

[Signature page, inventor declarations, and formal drawings to be appended per USPTO requirements prior to filing.]